



2 The continuous random variable X has probability density function f given by

$$f(x) = \begin{cases} \frac{4}{9}(x+1) & 0 \leq x \leq 1, \\ (x-2)^2 & 1 < x \leq 2, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find the cumulative distribution function of X . [3]

This image shows a full page of a handwriting practice worksheet. It consists of multiple sets of three horizontal dashed lines, providing a guide for letter height and placement. The lines are evenly spaced across the entire page, leaving ample room for writing practice. There is no text or other markings on the page.

(b) Find the exact value of the median of X . [2]





The random variable Y is defined by $Y = \sqrt{X}$.

- (c) Find the cumulative distribution function of Y . [3]

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- (d) Determine whether the median of Y is greater than, or less than, the median of X . [2]

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